

Midterm, Math 170S, Winter 2020
Instructor: Elizaveta Rebrova

Printed name: _____

Signed name: _____

Student ID number: _____

Instructions:

- Read problems very carefully. If you have any questions please ask.
- Do not forget to show all your work. The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems.
- Your final answers do not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- Books, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

α	0.400	0.300	0.200	0.100	0.050	0.025	0.020	0.010	0.005	0.001
z_α	0.253	0.524	0.842	1.282	1.645	1.960	2.054	2.326	2.576	3.090
$z_{\alpha/2}$	0.842	1.036	1.282	1.645	1.960	2.240	2.326	2.576	2.807	3.291

1. Let

- $X_1, X_2, \dots, X_{10}$ are samples taken from the distribution $N(\mu_X, 4)$, $\sum_{i=1}^{10} X_i = 5$,
- $Y_1, Y_2, \dots, Y_{15}$ are samples taken from the distribution $N(\mu_Y, 9)$, $\sum_{i=1}^{15} Y_i = 15$,
- $Z_1, Z_2, \dots, Z_8$ are taken from the distribution $N(\mu_Z, \sigma^2)$ distribution, $\sum_{i=1}^8 Z_i = 2$, we also know that their sample variance is 1.

All X_i 's, Y_i 's and Z_i 's are mutually independent.

(a) (3 points) Find 90% one-sided confidence interval for the upper bound for μ_X .

$$(-\infty; \bar{X} + z_{0.1}, \sqrt{\frac{s^2}{n}}] =$$

$$(-\infty; 0.5 + 1.282 \cdot \frac{2}{\sqrt{10}}]$$

(b) (3 points) Find 98% two-sided confidence interval for μ_Z .

$$\begin{aligned} & \left[\frac{1}{4} - \Phi_{0.01}(7) \cdot \frac{1}{\sqrt{8}}, \frac{1}{4} + \Phi_{0.01}(7) \cdot \frac{1}{\sqrt{8}} \right] \\ & = \left[\frac{1}{4} - 2.998 \cdot \frac{1}{\sqrt{8}}, \frac{1}{4} + 2.998 \cdot \frac{1}{\sqrt{8}} \right] \end{aligned}$$

(c) (2 points) What is the distribution of a random variable $\bar{Y} - 5\bar{X}$?

$$N\left(\mu_Y - 5\mu_X, \frac{9}{15} + \frac{4 \cdot 25}{10}\right) = N(-1.5, 10.4)$$

(d) (2 points) Find 95% two-sided confidence interval for a parameter $\mu_Y - 5\mu_X$.

$$P\left(-z_{0.025} \leq \frac{(\mu_Y - 5\mu_X) - (\bar{Y} - 5\bar{X})}{\sqrt{10.4}} \leq z_{0.025}\right) = 95\%$$

$$\mu_Y - 5\mu_X \in \left[-1.5 - z_{0.025} \cdot \sqrt{10.4}; -1.5 + z_{0.025} \cdot \sqrt{10.4}\right]$$

2. In a certain political campaign, more than 50% supportive votes will secure a win of a candidate. One candidate has a poll taken at random among the voting population, and computed confidence intervals (for the fraction of voters in favor) with confidence level 90%: a one-sided interval $[a, b]$ for the lower estimate; and a two-sided interval $[c, d]$.

(a) (2 points) What is b ?

$$b = 1$$

(b) (2 points) What can be bigger, a or c ?

$$a \geq c$$

(c) (3 points) What condition must be satisfied so that the candidate can be confident about her win with the confidence 90%?

$$a \geq \frac{1}{2} \quad (c > \frac{1}{2})$$

(d) (3 points) Suppose the candidate did extra polling and now the sample size increased four times ($n_{\text{new}} = 4n_{\text{old}}$). Will the length of the two-sided confidence interval increase or decrease? By how many times?

2 times

$$n = \frac{Z_{0.05}^2}{4 \epsilon^2}$$

$$n_{\text{new}} = 4n \Rightarrow$$

$$\epsilon_{\text{new}} = \frac{\epsilon}{2}$$

3. (10 points) We observe the following data sample

0.5, 0.3, 1, 0.5, 0.7

from the distribution with the density function

$$f(x) = \frac{1}{2\theta^3} x^2 e^{-x/\theta},$$

where $0 < x < \infty$ and unknown parameter θ is such that $0 < \theta < \infty$. Find Maximum Likelihood Estimate for θ .

$$f(x_1, \dots, x_5) = \frac{1}{(2\theta^3)^5} \prod x_i^2 \cdot e^{-\sum x_i / \theta}$$

$$l(\theta) = \log f(x_1, \dots, x_5) = -5 \log(2\theta^3) + 2 \sum \log x_i - \frac{\sum x_i}{\theta}$$

$$l'(\theta) = \frac{-5 \cdot 2 \cdot 3\theta^2}{2\theta^3} + \frac{\sum x_i}{\theta^2} = 0$$

$$-\frac{15}{\theta} + \frac{\sum x_i}{\theta^2} = 0$$

$$+ 15\theta = 0.5 + 0.3 + 1 + 0.5 + 0.7 = 3$$

$$\boxed{\hat{\theta} = \frac{1}{5}}$$

4. (a) (5 points) $X_1, \dots, X_n$ is a sample from the distribution $\text{Unif}[2\theta, 3\theta]$ depending on an unknown parameter $\theta > 0$. Find Method of Moments estimate for θ .

$$\bar{X} = E(\text{Unif}(2\theta, 3\theta)) = \frac{5}{2}\theta$$

$$\boxed{\hat{\theta} = \frac{2}{5} \bar{X}}$$

- (b) (5 points) Let $X_1, X_2, \dots, X_n$ are samples taken from $\text{Unif}[\theta, 5]$ depending on an unknown parameter $\theta < 5$. Using Factorization Theorem, show that the first order statistic of the data $Y_{(1)} = \min(X_1, \dots, X_n)$ is a sufficient statistic for θ .

$$f(x_1, \dots, x_n) = \frac{1}{(5-\theta)^n} \mathbb{1}_{\{\theta \leq x_1, \dots, x_n \leq 5\}}$$

$$= \frac{1}{(5-\theta)^n} \mathbb{1}_{\{\theta \leq Y_{(1)} \text{ and } x_1, \dots, x_n \leq 5\}}$$

$$= \underbrace{\frac{1}{(5-\theta)^n} \mathbb{1}_{\{\theta \leq Y_{(1)}\}}}_{h(\theta, Y_{(1)})} \cdot \underbrace{\mathbb{1}_{\{x_1, \dots, x_n \leq 5\}}}_{\varphi(x_1, \dots, x_n)}$$

Hint: recall that a convenient way to write density function for the uniform distribution on the segment $[a, b]$ is $f(x) = \frac{1}{b-a} \mathbb{1}_{\{a \leq x \leq b\}}$, where $\mathbb{1}$ denotes the indicator function (which is equal to one when $a \leq x \leq b$ and zero otherwise).

5. (10 points) We have 2 boxes, Box A and Box B, each box contains 3 balls. Box A contains 1 black and 2 white balls; Box B contains 2 black and 1 white ball. Our friend chooses one of the boxes at random, probability of choosing Box A is p ($0 < p < 1$). Then he takes one ball from a chosen box (each of three balls can be chosen equally likely), and it turns out to be white.

Using MAP estimate, for what values of p can we conclude that Box A was chosen? For what values of p can we conclude that Box B was chosen?

Prior: $\theta = \begin{cases} 0 & \text{with prob } p \\ 1 & \text{with prob } 1-p \end{cases}$

Conditional $P(\text{white ball} | \theta = 0) = \frac{2}{3}$ (2 white balls in Box A)

$P(\text{white ball} | \theta = 1) = \frac{1}{3}$

Posterior $P(\theta = 0 | \text{white Ball}) = \frac{p \cdot \frac{2}{3}}{(1-p) \cdot \frac{1}{3} + p \cdot \frac{2}{3}}$

$P(\theta = 1 | \text{white Ball}) = \frac{(1-p) \cdot \frac{1}{3}}{(1-p) \cdot \frac{1}{3} + p \cdot \frac{2}{3}}$

MAP $\hat{\theta} = 0$ if $\frac{2}{3}p > \frac{1}{3}(1-p)$

$\hat{\theta} = 1$ otherwise

If $p > \frac{1}{3} \Rightarrow \text{Box A}$, if $p < \frac{1}{3} \Rightarrow \text{Box B}$

Hint: Unknown parameter here is the box that was chosen (say, $\theta = 0$ if Box A was chosen and $\theta = 1$ if Box B was chosen). Find conditional distribution of observing a white ball, given θ (given box that was chosen). Find prior and posterior distributions of θ (they both will depend on p). Find the MAP estimate $\hat{\theta}$, which will give you the conclusion about which box was chosen. How does MAP estimate depend on p ?